week 1
↓
day 1 (task 2)

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time series

sequence of data points that occur in successive order over some period of time.

↳ taken on any variable that changes over time

Investing → track price of security over time

↳ on the hr over course of day

↳ at close on last day of every month over course of 5 yrs.

↳ how asset/ security/ economic variable change over time.

↳ good forecasts → genuine patterns/relationships existing in historical data but do not replicate past events that won't occur again.

↳ trend analysis, cyclic fluctuation, issues of seasonality

Models ⇒ (time forecasting)

1) decomposition
2) exponential smoothing
3) ARIMA

ex ⇒ stock prices, sales, data, temp

Time dependency matters

↳ past values influence future

forecast quality depends upon

↳ understanding structure of data

↳ identifying predictable vs random components

core components:

happens for some time and then disappears

a) trend

↳ long term movement (↑/↓)

→ ↑/↓/— (no trend ⇒ horizontal)

↳ non repeating (unlike seasonality)

↳ can change dir over time

b) seasonality → Statistical props not const

↳ regular, repeating patterns at fixed intervals (monthly, yearly)

↳ festival sales

weather cycles

↳ key prop ⇒ FIXED FREQUENCY

PREDICTABLE PATTERN

Why imp?
↳ help improve models by
↳ removing noise – clearer signal
↳ adding useful structured info

deseasonalization | Seasonal Adjustment
removing seasonal effects to simplify modeling

stationary time series

Seasonal component
- sine/cosine wave
- combination of periodic func over a generally fixed period & amplitude → approximate → curve fitting method

(c) cycles

↳ irregular fluctuations – not fixed freq

difference
1. NO FIXED PERIOD
2. USUALLY LONG DURATION
3. LESS PREDICTABLE

avg length longer
magnitude → more variable

d) Randomness (Irregular component)

↳ unpredictable variation
↳ can't be modeled effectively

Stationarity

statistical properties are constant over time

conditions:

↳ no trend
↳ no seasonality
↳ const mean & variance

if it has cycles then it can be non horizontal tt

↳ relationships (autocorrelation) stay same

Otherwise – its level should remain the same most of the time.

Why imp?
Most forecasting models require stationarity.

white noise

↳ special type of stationary tt.
↳ mean = 0
↳ const variance
↳ no autocorrelation (independent vals)

If variables are independent and identically distributed with a mean of 0.
this means ⇒ all variables hv the same variance ⇒ sigma²
& each value has a 0 correlation with all other values in the series

Importance ?

→ random

↳ Not predictable
↳ ideal model residuals should behave like white noise
↓↓
↳ means model captured all patterns
all signal info

error

GOAL
⇓
extract signal remove noise

left?
⇓ random fluctuations that can't be modelled